Arc Dipole Multipole Correction Procedure (corr_MB_v3.f)

Legacy code by S. Fartoukh (2010–2014)

1 Scope and Idea

This document explains the legacy Fortran procedure `corr_MB_v3.f` (LHC Project Report 501 style) that generates arc corrector settings to compensate field errors of the main dipoles (MB/MBH). The code:

- reads optics and error tables (`optics0_MB.mad`, `MB.errors`);
- converts integrated multipole errors into resonance-driving contributions using optics weights;
- solves for per-sector corrector strengths to cancel these contributions locally;
- distributes any remaining global skew terms using a pseudo-inverse;
- writes a MAD-X file `MB_corr_setting.mad`.

Eight arc sectors are used (periodicity of the LHC ring):

$$\text{sectors } s = 1 \ldots 8: \ (R1/L2,\ R2/L3,\ R3/L4,\ R4/L5,\ R5/L6,\ R6/L7,\ R7/L8,\ R8/L1).$$

Beam direction is inferred from the order of sectors in `optics0_MB.mad` and affects sign conventions.

Corrector families. MQS (skew quads), MSS (skew sexts), MCS (normal sext spool), MCO (normal oct spool), MCD (normal deca spool), and MQT (normal quad trims, focus/defocus). Lengths are denoted l.MQS, l.MSS, etc. Integrated strengths printed in `MB_corr_setting.mad` are divided by these lengths.

2 Optics Weighting

For each MB slice the optics file provides $\beta_x, \beta_y, D_x, \Delta\mu_x, \Delta\mu_y$. Define

$$\Delta\mu = 2\pi(\Delta\mu_x - \Delta\mu_y), \qquad c = \cos\Delta\mu, \qquad s = \sin\Delta\mu.$$

Store per slice and sector s:

$$\begin{aligned}
&\text{b2 weights:} \quad w_{b2,x} = \beta_x, \quad w_{b2,y} = \beta_y, \\
&\text{a2 weights:} \quad w_{a2,c} = \sqrt{\beta_x\beta_y}\, c, \quad w_{a2,s} = \sqrt{\beta_x\beta_y}\, s, \\
&\text{a3 weights:} \quad w_{a3,c} = \sqrt{\beta_x\beta_y}\, D_x\, c, \quad w_{a3,s} = \sqrt{\beta_x\beta_y}\, D_x\, s.
\end{aligned}$$

Corrector responses are accumulated similarly:

- MQS (skew quad): $a2_c(s) = \sum w_{a2}/(2\pi) \cdot \ell_{\mathrm{MQS}}$.

- MSS (skew sext): $a3_c(s) = \sum w_{a3}/(2\pi)$.
- MQT focus/defocus (normal quad trims): $b2_c^{F/D}(s) = \sum(\beta_x/2,\ -\beta_y/2)/(2\pi) \cdot \ell_{\mathrm{MQT}}$, with polarity chosen from sector parity, magnet index, and beam.

After counting slices, responses are renormalized (average over all corrector slices): $a2_c \leftarrow a2_c \cdot 4/N_{\mathrm{MQS}}$, $b2_c^{F/D} \leftarrow b2_c^{F/D} \cdot 8/N_{\mathrm{MQTF/D}}$; baseline strengths $k_{\mathrm{QTF/QTD}}$ are scaled accordingly.

3 Error Aggregation

From `MB.errors` each MB slice provides integrated multipoles $K_n, K_n^{(\mathrm{sk})}$. Per sector s accumulate

$$
\begin{aligned}
b2_b &= \sum \big(w_{b2,x} K_1 - w_{b2,y} K_1\big)/(2\pi), \\
a2_b &= \sum w_{a2}\, K_1^{(\mathrm{sk})}/(2\pi), \\
a3_b &= \sum w_{a3}\, K_2^{(\mathrm{sk})}/(2\pi), \\
b3_b &= \sum K_2, \qquad b4_b = \sum K_3, \qquad b5_b = \sum K_4.
\end{aligned}
$$

Counts of MCS/MCO/MCD spools are stored to average later.

4 First (Local) Correction

Normal quad (b2). Each sector uses the two MQT families (focus/defocus) to solve

$$
\begin{bmatrix} b2_c^{F,x} & b2_c^{D,x} \\ b2_c^{F,y} & b2_c^{D,y} \end{bmatrix} \begin{bmatrix} k_F \\ k_D \end{bmatrix} = -\begin{bmatrix} b2_b^x \\ b2_b^y \end{bmatrix}, \qquad (k_F, k_D) \to b2(s).
$$

Skew quad (a2). There is one skew-quad knob per sector, but the error has two phase components (cosine/sine). Model it as a 2D vector equation

$$
a2_{\mathrm{loc}}(s)\, \mathbf{a2}_c(s) + \mathbf{a2}_b(s) \approx \mathbf{0}, \quad \mathbf{a2}_c(s) = \big(a2_c^1(s),\, a2_c^2(s)\big),\ \mathbf{a2}_b(s) = \big(a2_b^1(s),\, a2_b^2(s)\big).
$$

The best scalar $a2_{\mathrm{loc}}(s)$ that cancels along $\mathbf{a2}_c$ is the projection of $-\mathbf{a2}_b$ on $\mathbf{a2}_c$:

$$
a2_{\mathrm{loc}}(s) = -\frac{\mathbf{a2}_c \cdot \mathbf{a2}_b}{\|\mathbf{a2}_c\|^2} = -\frac{\sum_i a2_c^i(s)\, a2_b^i(s)}{\sum_i (a2_c^i(s))^2}.
$$

Whatever is orthogonal to $\mathbf{a2}_c$ cannot be corrected locally; the remaining 2D error is accumulated as a residual for the global step:

$$
a2_{\mathrm{res}}^i \mathrel{+}= a2_{\mathrm{loc}}(s)\, a2_c^i(s) + a2_b^i(s).
$$

Net effect: kill the component parallel to the local response, carry the perpendicular component forward.

Skew sext (a3). Same projection, but with the $a3$ weights (which include D_x):

$$
a3_{\mathrm{loc}}(s) = -\frac{\sum_i a3_c^i(s)\, a3_b^i(s)}{\sum_i (a3_c^i(s))^2}, \qquad a3_{\mathrm{res}}^i \mathrel{+}= a3_{\mathrm{loc}}(s)\, a3_c^i(s) + a3_b^i(s).
$$

Normal sext/oct/deca. Simple averages over spools:

$$b3(s) = -\frac{b3_b(s)}{N_{\mathrm{MCS}}}, \quad b4(s) = -\frac{b4_b(s)}{N_{\mathrm{MCO}}}, \quad b5(s) = -\frac{b5_b(s)}{N_{\mathrm{MCD}}}.$$

At collision energy, magnitudes are clipped to hardware limits $k_{\mathrm{MCS}}^{\max}, k_{\mathrm{MCO}}^{\max}, k_{\mathrm{MCD}}^{\max}$.

5 Second (Global) Correction for a2 and a3

After the local step, each sector still leaves a 2D residual vector (cos/sin) for skew quads and skew sexts:

$$\mathbf{a2}_{\mathrm{res}} = \left(a2^1_{\mathrm{res}}, a2^2_{\mathrm{res}}\right), \qquad \mathbf{a3}_{\mathrm{res}} = \left(a3^1_{\mathrm{res}}, a3^2_{\mathrm{res}}\right).$$

Goal: distribute these two global components across all 8 sector knobs so that, when added up, they cancel the residual vector in a least-squares sense.

Response matrix. Collect the sector responses into an 8×2 matrix

$$A = \begin{bmatrix} a2^1_c(1) & a2^2_c(1) \\ \vdots & \vdots \\ a2^1_c(8) & a2^2_c(8) \end{bmatrix}.$$

Each row is the two phase components of the skew-quad response in one sector.

Pseudo-inverse. The code builds

$$B = -A^\top (AA^\top)^{-1}.$$

This is the (negative) Moore–Penrose left pseudo-inverse of A; multiplying a 2D vector by B yields an 8D vector of sector weights whose projection through A best cancels that 2D vector in the least-squares sense.

Then the final per-sector strengths are

$$a2(s) = a2_{\mathrm{loc}}(s) + \sum_{j=1}^{2} B_{sj}\, a2^j_{\mathrm{res}}, \qquad a3(s) = a3_{\mathrm{loc}}(s) + \sum_{j=1}^{2} B_{sj}\, a3^j_{\mathrm{res}}.$$

Interpretation:

- $a2_{\mathrm{loc}}(s)$ (or $a3_{\mathrm{loc}}$) is what the sector did locally.
- $B_{sj} a2^j_{\mathrm{res}}$ is that sector's share of the global residual component j, chosen so that the weighted sum of all sectors reproduces (and cancels) the residual vector as closely as possible.
- Because there are 8 knobs but only 2 components to cancel, this distributes the correction smoothly across sectors rather than loading a single sector.

6 Outputs

- `MB_corr_setting.mad` with:
  - Quadrupole trims: per-sector ΔKQTF/KQTD and final KQTF/KQTD including baseline and beam sign.
  - Skew quad (KQS), skew sext (KSS), normal sext (KCS), normal oct (KCO), normal deca (KCD) per sector, divided by element lengths.
- Sign conventions: depend on beam (B1 vs B2; a special ibeam=4 case if beam 2 and reversed order). Beam direction flag bv flips certain families; the sign choice matches the MAD-X convention for the two beams.
- Saturation: at collision energy, spools are clipped to hardware limits ($k^{\mathrm{max}}_{\mathrm{MCS/MCO/MCD}}$) before printing.

7 Reference Run

Reference inputs are `temp/optics0_MB.mad` and `temp/MB.errors`; the legacy output is `temp/MB_corr_setting.mad` in the same folder, matching the method above and the strategy described in LHC Project Report 501.

Reading tip. If you want to trace a specific term: (i) find its weight in the optics section, (ii) see how it is multiplied by the corresponding K_n column in `MB.errors`, (iii) note which corrector family and sector solve it (local) or receive a share via the pseudo-inverse (global skew terms).